Pavithra Uthrah R. K.

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Profile

Data Engineer who builds tested, observable data pipelines on Azure Databricks. At KaarTech, built a PySpark ETL pipeline into a Power BI star schema with schema contracts, data quality checks and 39 automated tests, and Sentinel, a Databricks App for operating SAP-sourced pipelines with incident workflows and data validation. Strong in SQL, Python, PySpark, Delta Lake, medallion architecture and dimensional modeling. Chairperson of a 250+ member technical community. Seeking a full-time Data Engineer role starting Jan 2027.

Experience

Data Engineer Intern – KaarTech, Chennai Internship | On-site | Jul 2026 – Oct 2026

- Built a PySpark ETL pipeline converting raw FMCG sales exports into a Power BI-ready star schema (1 fact, 6 dimension tables), with an explicit schema contract, configurable data quality thresholds and dimension-key uniqueness checks before every join to prevent row explosion, backed by 39 automated pytest tests.
- Normalized placeholder tokens ("NA", "-") and comma-formatted numbers to NULL before casting, and redesigned the export stage so the pipeline runs on Windows, Linux and macOS with a single command.
- Built **Sentinel**, a pipeline operations console for SAP-sourced Databricks pipelines, deployed as a Databricks App; models 15 pipelines across 5 countries and 40 runs a day with an 8-step incident workflow (investigation, failure classification, SLA, remediation, validation, notification, regression test) covering 14 failure types.
- Designed the incident rules: safe-to-rerun failures auto-remediate, other recoverable failures wait for approval, and unfixable ones escalate by email with a full audit history; runs slower than the 15-minute average are flagged Critical.
- Built a data validation module running 10 data quality checks across 7 source and result tables in Unity Catalog, and a live Genie assistant behind a server-side proxy that authenticates as a service principal, so no Databricks credential reaches the browser.

Software Development Engineer Intern – AIC Crescent Innovation & Incubation Council, Chennai
Internship | On-site | May 2026 – Jun 2026

- Built a FastAPI pipeline that parses unstructured WhatsApp worklogs into structured task records in PostgreSQL and Google Sheets, reconciling task state to carry forward pending work automatically.
- Designed the PostgreSQL schema and Django backend for a facility booking system serving 140+ incubated startups, with slot scheduling, conflict detection and automated billing; one of 3 backend systems delivered in a 1-month internship.
- Modeled a multi-tenant ERP data foundation with role-based access control across 14 entities, designed for extensibility.

Projects

EV Fleet Lakehouse Platform *Databricks, PySpark, Delta Lake, Unity Catalog, Auto Loader, Power BI*

- Built a Bronze/Silver/Gold lakehouse integrating 6 source systems (telematics, fleet, trips, charging, diagnostics, maintenance), with Auto Loader incremental ingestion and idempotent MERGE-based Silver loads that deduplicate and validate records.
- Published 30 fleet KPIs across 5 domains to Power BI from Gold tables; a Delta audit table records per-run record counts, rejected rows and quality status, with Unity Catalog governance.

Signal360 – Customer 360 and CDC Lakehouse *PySpark, Delta Lake, Airflow, pytest, GitHub Actions*

- Built a Customer 360 lakehouse that applies insert, update and delete change events with idempotent Delta MERGE and keeps plan history as SCD Type 2, so churn analysis reflects the plan a customer held at the time.
- Implemented entity resolution (exact and fuzzy matching) to merge duplicate customer records into golden records; engineered churn features with window functions; orchestrated with Airflow.
- Quarantined invalid change records, and covered merge and SCD2 logic with pytest running in CI.

Real-Time Sensor Streaming Pipeline *Kafka, Spark Structured Streaming, Delta Lake, Docker*

- Built a streaming pipeline that replays IoT sensor telemetry through Kafka into Spark Structured Streaming and Delta bronze and silver tables, using watermarking, checkpointing and deduplication to handle late and duplicate events.
- Computed 5-minute windowed aggregates and threshold-based anomaly alerts; packaged with Docker Compose for one-command local deployment.

Skills

- **Languages:** SQL, Python, Java, C/C++
- **Data Engineering:** PySpark, Azure Databricks, Delta Lake, Unity Catalog, Medallion Architecture, ETL/ELT, Dimensional Modeling (Star Schema), Data Quality Validation, Auto Loader
- **Analytics & BI:** Power BI, DAX, Tableau, Pandas, NumPy
- **Databases & APIs:** PostgreSQL, MySQL, Query Optimization, FastAPI, Django, REST APIs

- **Tools & Cloud:** Microsoft Azure, Git/GitHub, pytest, Docker, CI/CD, Linux
- **Core CS:** Data Structures & Algorithms, DBMS, OOP, Operating Systems, Computer Networks

Education

Bachelor of Technology in Computer Science and Engineering Aug 2023 – May 2027
Vellore Institute of Technology, Chennai

Higher Secondary Education (Class XII) Graduated May 2023
ISC — St. Michael's Academy, Chennai

Leadership & Activities

- **Chairperson, Open Source Programming Club, VIT Chennai** (Apr 2025 – Present)
  - Lead a 250+ member technical community across operations, partnerships and cross-functional teams; organized 3 national-level hackathons and 10+ technical events.
- **1st Place – HackHub'25** (IEEE CompSoc VIT-C, GitHub & Devfolio), among 500 teams, for an AI-driven autism screening tool built in 24 hours.
- **Organizer – Spectrum, VOID.v1, Glytch, TechnoVIT'25, VIT Chennai**: coordinated operations and logistics for technical festivals with 3000+ participants.
- **Student Mentor Volunteer, U&I Trust, India** (Jul 2026 – Present): weekly academic tutoring and mentorship for primary school students.

Certifications

- **Google Data Analytics Professional Certificate (v.3)** Issued by Google, January 2026.
- **Data Science & Analytics Certificate** Issued by HP LIFE, August 2026.